K-Means

Objetivo:
$\min_{\{r_{nk}\},\{\boldsymbol{\mu}_k\}} J=\sum_n\sum_k r_{nk}\|\mathbf{x}_n-\boldsymbol{\mu}_k\|^2$
$r_{nk}\in\{0,1\}$, $\sum_k r_{nk}=1$ (asignación *hard*)

Inicialización $\boldsymbol{\mu}_k$ (random o K-Means++)
$\approx$**E-Step**: $z_n^*\leftarrow\arg\min_k\|\mathbf{x}_n-\boldsymbol{\mu}_k\|^2$
$\approx$**M-Step**: $\boldsymbol{\mu}_k\leftarrow\frac{1}{N_k}\sum_{n:\,z_n^*=k}\mathbf{x}_n$
K-Means++: $p(\boldsymbol{\mu}_{t+1}=\mathbf{x}_n)=\frac{d_t(\mathbf{x}_n)}{\sum_{n'}d_t(\mathbf{x}_{n'})}$
$d_t(\mathbf{x})=\min_{k=1}^t\|\mathbf{x}-\boldsymbol{\mu}_k\|_2^2$
Límite de GMM: $\boldsymbol{\Sigma}_k=\varepsilon\mathbf{I}$, $\varepsilon\to0\Rightarrow$ asignación hard (gaussianas infinitamente concentradas).

Gaussian Mixture Model (GMM)

$p(\mathbf{x}|\mathbf{w})=\sum_{k=1}^K\pi_k\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k)$; $\sum_k\pi_k=1$
Latente: $z\sim\mathrm{Cat}(\boldsymbol{\pi})$; $\mathbf{x}|z=k\sim\mathcal{N}(\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k)$
$\ln p(\mathbf{X}|\mathbf{w})=\sum_n\ln\sum_k\pi_k\mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k)$

E-step (responsabilidades, soft clust.):
$$r_{nk}=p(\mathbf{z}=k\mid x_n,\boldsymbol{\theta})=\frac{\pi_k\mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k)}{\sum_{k'}\pi_{k'}\mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_{k'},\boldsymbol{\Sigma}_{k'})}$$
M-step:
$N_k=\sum_n r_{nk}$; $\hat{\pi}_k=\frac{N_k}{N}$; $\hat{\boldsymbol{\mu}}_k=\frac{1}{N_k}\sum_n r_{nk}\mathbf{x}_n$
$\hat{\boldsymbol{\Sigma}}_k=\frac{1}{N_k}\sum_n r_{nk}(\mathbf{x}_n-\hat{\boldsymbol{\mu}}_k)(\mathbf{x}_n-\hat{\boldsymbol{\mu}}_k)^\top$
Asignación (hard clust.): $\mathbf{z}_n=\arg\max_k r_{nk}$
Frontera GMM: $r_{nk}=r_{nk'}$, lineal si $\boldsymbol{\Sigma}_k=\boldsymbol{\Sigma}\forall k$.

EM & ELBO

Goal: $\max_{\mathbf{w}}\ln p(\mathbf{x}|\mathbf{w})=\max_{\mathbf{w}}\ln\sum_{\mathbf{z}}p(\mathbf{x},\mathbf{z}|\mathbf{w})$
Asumiendo $q(\mathbf{z})$:
$$\ln p(\mathbf{x}|\mathbf{w})=\underbrace{\mathbb{E}_{q(\mathbf{z})}\left[\ln\frac{p(\mathbf{x},\mathbf{z}|\mathbf{w})}{q(\mathbf{z})}\right]}_{Ł(\mathbf{w},q)}+\underbrace{D_{\mathrm{KL}}(q(\mathbf{z})\|p(\mathbf{z}|\mathbf{x},\mathbf{w}))}_{\geq0}$$
$\ln p(\mathbf{x}|\mathbf{w})\geq Ł(\mathbf{w},q)$
E-step: Maximizar Ł respecto a q
$q^*=p(\mathbf{z}|\mathbf{x},\mathbf{w}^{t-1})\Rightarrow$ KL$=0$, Ł $=\ln p$
M-step: Maximizar Ł respecto a $\mathbf{w}$
$\mathbf{w}^t=\arg\max_{\mathbf{w}}\mathbb{E}_{q(\mathbf{z})}[\ln p(\mathbf{x},\mathbf{z}|\mathbf{w})]$
Convergencia: $\ln p(\mathbf{x}|\mathbf{w}^t)\geq\ln p(\mathbf{x}|\mathbf{w}^{t-1})$
(no-decreciente; converge al menos a máx. local)
ELBO expandido: Ł $=\mathbb{E}_q[\ln p(\mathbf{x},\mathbf{z}|\mathbf{w})]+\mathbb{H}(q)$

Backpropagation

Forward ($l=1..L$): $\mathbf{a}^l=\mathbf{W}^l\mathbf{z}^{l-1}+\mathbf{b}^l$;
$\mathbf{z}^l=h(\mathbf{a}^l)$; $\mathbf{z}^0=\mathbf{x}$
Error de salida ($\boldsymbol{\delta}^L\equiv\partial\mathcal{L}/\partial\mathbf{a}^L$):
$\boldsymbol{\delta}^L=\nabla_{\mathbf{z}^L}\mathcal{L}\odot h'(\mathbf{a}^L)$
MSE+lineal o BCE+sigmoide $\Rightarrow\boldsymbol{\delta}^L=\mathbf{z}^L-\mathbf{y}$
Backward ($l=L-1..1$):
$\boldsymbol{\delta}^l=(\mathbf{W}^{l+1\top}\boldsymbol{\delta}^{l+1})\odot h'(\mathbf{a}^l)$
Gradientes: $\nabla_{\mathbf{W}^l}\mathcal{L}=\boldsymbol{\delta}^l\mathbf{z}^{l-1\top}$; $\nabla_{\mathbf{b}^l}\mathcal{L}=\boldsymbol{\delta}^l$
Update: $\mathbf{W}^l\leftarrow\mathbf{W}^l-\eta\nabla_{\mathbf{W}^l}\mathcal{L}$
Derivadas usuales: $\sigma'=\sigma(1-\sigma)$;
$\tanh'=1-\tanh^2$; $\mathrm{ReLU}'=\mathbb{I}[\mathbf{a}>0]$

Demo: Convergencia de EM

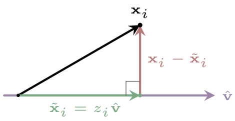
Una iteración da $\ln p(\mathbf{x}|\mathbf{w}^t)\geq\ln p(\mathbf{x}|\mathbf{w}^{t-1})$
(1) E-step: $q^*=p(\mathbf{z}|\mathbf{x},\mathbf{w}^{t-1})\Rightarrow$ KL $=0$:
$Ł(\mathbf{w}^{t-1},q^*)=\ln p(\mathbf{x}|\mathbf{w}^{t-1})$
(2) M-step: $\mathbf{w}^t=\arg\max_{\mathbf{w}}Ł(\mathbf{w},q^*)$:
$Ł(\mathbf{w}^t,q^*)\geq Ł(\mathbf{w}^{t-1},q^*)$ (def. de máximo)
(3) Cota inferior (vale $\forall\mathbf{w}$, KL≥0):
$\ln p(\mathbf{x}|\mathbf{w}^t)\geq Ł(\mathbf{w}^t,q^*)$
Encadenando:
$\ln p(\mathbf{x}|\mathbf{w}^t)\overset{(3)}{\geq}Ł(\mathbf{w}^t,q^*)\overset{(2)}{\geq}Ł(\mathbf{w}^{t-1},q^*)\overset{(1)}{=}$
$\ln p(\mathbf{x}|\mathbf{w}^{t-1})$
$\Rightarrow\ln p(\mathbf{x}|\mathbf{w}^t)\geq\ln p(\mathbf{x}|\mathbf{w}^{t-1})$. $\square$
Max local: secuencia monótona $\uparrow$ y acotada $\Rightarrow$ converge, pero a un punto estacionario (no global, pues $\ln p$ es no cóncava; depende de $\mathbf{w}^0$).

Demo: $\mathcal{L}$ VAE

$$\ln p(\mathbf{x}|\theta,\phi)=\ln\int p_\theta(\mathbf{x},\mathbf{z})d\mathbf{z}=\ln\int q_\phi(\mathbf{z}|\mathbf{x})\frac{p_\theta(\mathbf{x},\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}$$
$$=\ln\mathbb{E}_{q_\phi}\left[\frac{p_\theta(\mathbf{x},\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right]$$
$$\text{(Jensen)}\geq\mathbb{E}_{q_\phi}\left[\ln\frac{p_\theta(\mathbf{x},\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right]$$
$$=\mathbb{E}_{q_\phi}[\ln p_\theta(\mathbf{x}|\mathbf{z})+\ln p(\mathbf{z})-\ln q_\phi(\mathbf{z}|\mathbf{x})]$$
$$=\mathbb{E}_{q_\phi}[\ln p_\theta(\mathbf{x}|\mathbf{z})]-\mathbb{E}_{q_\phi}\left[\ln\frac{q_\phi(\mathbf{z}|\mathbf{x})}{p(\mathbf{z})}\right]$$
$$=\mathbb{E}_{q_\phi}[\ln p_\theta(\mathbf{x}|\mathbf{z})]-D_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}))$$

PCA

Centrar: $\tilde{\mathbf{X}}$; $\hat{\boldsymbol{\Sigma}}=\frac{1}{N}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}$
Eigendecomposición: $\hat{\boldsymbol{\Sigma}}\mathbf{u}_l=\lambda_l\mathbf{u}_l$;
$\mathbf{U}_L=[\mathbf{u}_1,\ldots,\mathbf{u}_L]\in\mathbb{R}^{D\times L}$
Encode: $\mathbf{z}_n=\mathbf{U}_L^\top\tilde{\mathbf{x}}_n$; **Decode**: $\hat{\mathbf{x}}_n=\mathbf{U}_L\mathbf{z}_n+\hat{\boldsymbol{\mu}}$
Error recons.: $\frac{1}{N}\|\tilde{\mathbf{X}}-\tilde{\mathbf{X}}\mathbf{U}_L\mathbf{U}_L^\top\|_F^2=\sum_{l=L+1}^D\lambda_l$
Var. explicada: $\mathrm{VE}(L)=(\sum_{l=1}^L\lambda_l)/(\sum_{l=1}^D\lambda_l)$
Vía SVD: $\tilde{\mathbf{X}}=\mathbf{U}_{\mathbf{X}}\mathbf{S}\mathbf{V}^\top$; PC's = cols de $\mathbf{V}$;
$\lambda_l=s_l^2/N$
Var de z: $\mathrm{Var}[z_l]=\lambda_l$ (coordenadas decorreladas)



Autoencoders

AE determinístico: $\mathbf{z}=f_\phi(\mathbf{x})$, $\hat{\mathbf{x}}=g_\theta(\mathbf{z})$, $L\ll D$
$\mathcal{L}_{\mathrm{AE}}=\frac{1}{N}\sum_n\|\mathbf{x}_n-g_\theta(f_\phi(\mathbf{x}_n))\|^2$
AE lineal $\equiv$ PCA (mismo subespacio); con no-linealidades $\to$ reducción no-lineal.

VAE: prior $p(\mathbf{z})=\mathcal{N}(\mathbf{0},\mathbf{I})$
Encoder: $q_\phi(\mathbf{z}|\mathbf{x})=\mathcal{N}(\mathbf{z}|\boldsymbol{\mu}_\phi(\mathbf{x}),\mathrm{diag}(\boldsymbol{\sigma}_\phi^2(\mathbf{x})))$
Decoder: $p_\theta(\mathbf{x}|\mathbf{z})=\mathcal{N}(\mathbf{x}|f_d(\mathbf{z};\theta),\sigma^2\mathbf{I})$
ELBO (a maximizar):
$\mathcal{L}_{\mathrm{VAE}}=\mathbb{E}_{q_\phi}[\ln p_\theta(\mathbf{x}|\mathbf{z})]-\mathrm{KL}(q_\phi(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}))$
KL gaussiano:
$\mathrm{KL}(\mathcal{N}(\boldsymbol{\mu},\mathrm{diag}(\boldsymbol{\sigma}^2))\|\mathcal{N}(\mathbf{0},\mathbf{I}))$
$=\frac{1}{2}\sum_j(\mu_j^2+\sigma_j^2-\ln\sigma_j^2-1)$
Reparam.: $\mathbf{z}=\boldsymbol{\mu}_\phi(\mathbf{x})+\boldsymbol{\sigma}_\phi(\mathbf{x})\odot\boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon}\sim\mathcal{N}(\mathbf{0},\mathbf{I})$
Forward: $\mathbf{x}\to f_e\to\underset{\mathcal{L}_{KL}}{(\boldsymbol{\mu},\boldsymbol{\sigma})}\to\mathbf{z}\to f_d\to\underset{\mathcal{L}_{\mathrm{rec}}}{\hat{\mathbf{x}}}$
Backward: $\overset{\mathcal{L}_{\mathrm{rec}}}{\nabla_\theta f_d}\to\overset{\mathcal{L}_{KL}}{\nabla(\boldsymbol{\mu},\boldsymbol{\sigma})}\to\nabla_\phi f_e\to\nabla\mathcal{L}$
$$\nabla\mathcal{L}\begin{cases}\nabla_\theta\mathcal{L}=\nabla_\theta\mathcal{L}_{\mathrm{rec}}\\\nabla_\phi\mathcal{L}=\nabla_\phi\mathcal{L}_{\mathrm{rec}}+\nabla_\phi\mathcal{L}_{\mathrm{KL}}\end{cases}$$

Information Theory

Entropía: $\mathbb{H}(p)=-\sum_i p_i\log p_i\geq0$
KL divergence (no simétrica, ≥0):
$D_{\mathrm{KL}}(q\|p)=\sum_i q_i\ln\frac{q_i}{p_i}=\mathbb{E}_q\left[\ln\frac{q}{p}\right]$
$D_{\mathrm{KL}}(q\|p)=0\Leftrightarrow q=p$; $D_{\mathrm{KL}}(q\|p)\neq D_{\mathrm{KL}}(p\|q)$
KL gaussianas ($p=\mathcal{N}(\boldsymbol{\mu}_1,\boldsymbol{\Sigma}_1)$, $q=\mathcal{N}(\boldsymbol{\mu}_0,\boldsymbol{\Sigma}_0)$):
$$D_{\mathrm{KL}}(q\|p)=\tfrac{1}{2}\Big[\ln\frac{|\boldsymbol{\Sigma}_1|}{|\boldsymbol{\Sigma}_0|}+\mathrm{tr}(\boldsymbol{\Sigma}_1^{-1}\boldsymbol{\Sigma}_0)+(\boldsymbol{\mu}_1-\boldsymbol{\mu}_0)^\top\boldsymbol{\Sigma}_1^{-1}(\boldsymbol{\mu}_1-\boldsymbol{\mu}_0)-D\Big]$$
Caso $p=\mathcal{N}(\mathbf{0},\mathbf{I})$:
$D_{\mathrm{KL}}=\frac{1}{2}\sum_j(\mu_j^2+\sigma_j^2-\ln\sigma_j^2-1)$

Métricas de Clustering

$\underset{\text{within}}{\mathrm{WCSS}}=\sum_k\sum_{n:r_{nk}=1}\|\mathbf{x}_n-\boldsymbol{\mu}_k\|^2$ ($\downarrow$ mejor)
$\underset{\text{between}}{\mathrm{BCSS}}=\sum_k N_k\|\boldsymbol{\mu}_k-\bar{\mathbf{x}}\|^2$ ($\uparrow$ mejor)
Silhouette $s(n)\in[-1,1]$: $a(n)$ = dist. intra, $b(n)$ = dist. al vecino más cercano
$s(n)=\frac{b(n)-a(n)}{\max(a(n),b(n))}$; $\bar{s}=\frac{1}{N}\sum_n s(n)$
Davies-Bouldin: $\mathrm{DB}=\frac{1}{K}\sum_k\max_{k'\neq k}\frac{s_k+s_{k'}}{d(\boldsymbol{\mu}_k,\boldsymbol{\mu}_{k'})}$
Elbow: graficar WCSS vs. K; codo = K óptimo.
BIC: $-2\ln\hat{p}(\mathbf{X})+d_K\ln N$; $\downarrow$ mejor; selecciona K.
Mutual. Info (con labels):
$$I(S_k;Y)=\sum_i\sum_j p_{C_iY}(i,j)\log\frac{p_{C_iY}(i,j)}{p_{C_i}(i)\,p_Y(j)}$$
$p_{C_iY}(i,j)=\frac{|c_i\cap y_j|}{N}$; $I(X;Y)=0\Leftrightarrow X\perp Y$
$I(X;Y)=\mathbb{H}(X)-\mathbb{H}(X|Y)=\mathbb{H}(Y)-\mathbb{H}(Y|X)$
NMI (Normalized Mutual Information):
$$\mathrm{NMI}(S_k,Y)=\frac{2\,I(S_k;Y)}{\mathbb{H}(S_k)+\mathbb{H}(Y)}\in[0,1]\quad(\uparrow\text{ mejor})$$

Demo: PCA invariante roto-traslaciones

$\mathbf{x}_i'=R\mathbf{x}_i+\mathbf{b}$
Traslación: $\mathbf{x}_i'=\mathbf{x}+\mathbf{b}\implies\boldsymbol{\mu}'=\boldsymbol{\mu}+\mathbf{b}$.
$\tilde{\mathbf{x}}=\mathbf{x}_i'-\boldsymbol{\mu}'=\mathbf{x}_i+\mathbf{b}-(\boldsymbol{\mu}+\mathbf{b})=\mathbf{x}_i-\boldsymbol{\mu}$.
Rotación: $\tilde{\mathbf{x}}_i'=\mathbf{R}\tilde{\mathbf{x}}_i$, $\mathbf{R}$ de rotación ($\perp$, det $=1$).
$\hat{\boldsymbol{\Sigma}}'=\mathbf{R}\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top\mathbf{R}^\top=\mathbf{R}\hat{\boldsymbol{\Sigma}}\mathbf{R}^\top$
Si $\hat{\boldsymbol{\Sigma}}\mathbf{u}_l=\lambda_l\mathbf{u}_l\Rightarrow\hat{\boldsymbol{\Sigma}}'(\mathbf{R}\mathbf{u}_l)=\lambda_l(\mathbf{R}\mathbf{u}_l)$
Coords reducidas:
$z_l'=(\mathbf{R}\mathbf{u}_l)^\top(\mathbf{R}\tilde{\mathbf{x}}_i)=\mathbf{u}_l^\top\mathbf{R}^\top\mathbf{R}\tilde{\mathbf{x}}_i=z_l$.

Demo: PCA: $\max\mathbb{V}[\mathbf{z}]\equiv\min\frac{1}{N}\|\mathbf{x}-\tilde{\mathbf{x}}\|^2$

$\bar{\mathbf{z}}=\frac{1}{N}\sum z_i=\bar{\mathbf{x}}^T\hat{v}=0\implies\mathbb{V}[\mathbf{z}]=\frac{1}{N}\sum z_i^2$
$$\min_{\hat{v}}\frac{1}{N}\sum\|x_i-z_i\hat{v}\|^2\equiv\max_{\hat{v}}\frac{1}{N}\sum z_i^2$$
$$\min_{\hat{v}}\frac{1}{N}\sum\|x_i\|^2-z_i^2\equiv\max_{\hat{v}}\frac{1}{N}\sum z_i^2$$
$$\min_{\hat{v}}-\frac{1}{N}\sum z_i^2\equiv\max_{\hat{v}}\frac{1}{N}\sum z_i^2$$

Clustering+

HAC (Hierarchical Agglomerative):
Bottom-up: empieza con N clusters, fusiona iterativamente los más cercanos $\to$ dendrograma.
No requiere K a priori.
Linkage: single (min dist), complete (max dist), average, Ward (min ΔWCSS).
DBSCAN: params ε, MinPts.
Core: $|\mathcal{N}_\varepsilon(\mathbf{x})|\geq$ MinPts; *Border*: vecino de core; *Noise*: resto.
Clusters = componentes conexas de cores + borders.

for $\mathbf{x}_n$ no visitado **do**
  if $|\mathcal{N}_\varepsilon(\mathbf{x}_n)|\geq$ MinPts **then** nuevo cluster; expandir($\mathbf{x}_n$)
  else noise

No requiere K; detecta formas arbitrarias; sensible a ε.
Spectral Clustering:
Grafo de similitud $\mathbf{W}$; Laplaciano $\mathbf{L}=\mathbf{D}-\mathbf{W}$.
Usar los K autovectores de menor λ de $\mathbf{L}$ como features; aplicar K-Means. Detecta clusters no convexos.
Mixture of Bernoullis ($\mathbf{x}\in\{0,1\}^D$):
$p(\mathbf{x}|\mathbf{w})=\sum_k\pi_k\prod_d\mu_{kd}^{x_d}(1-\mu_{kd})^{1-x_d}$
EM igual que GMM; M-step: $\hat{\mu}_{kd}=\frac{\sum_n r_{nk}x_{nd}}{N_k}$

Álgebra Lineal

Traza y ciclo: $\mathrm{tr}(\mathbf{ABC})=\mathrm{tr}(\mathbf{CAB})=\mathrm{tr}(\mathbf{BCA})$
$\mathbf{x}^\top\mathbf{A}\mathbf{x}=\mathrm{tr}(\mathbf{A}\mathbf{x}\mathbf{x}^\top)$ (escalar $\to$ traza)
Gradientes matriciales clave:
$\nabla_{\mathbf{A}}\mathrm{tr}(\mathbf{AB})=\mathbf{B}^\top$
$\nabla_{\mathbf{A}}\ln|\mathbf{A}|=\mathbf{A}^{-\top}$ (i.e. $\boldsymbol{\Sigma}^{-1}$ si $\boldsymbol{\Sigma}$ SDP)
$\nabla_{\mathbf{A}}(\mathbf{x}^\top\mathbf{A}^{-1}\mathbf{x})=-\mathbf{A}^{-\top}\mathbf{x}\mathbf{x}^\top\mathbf{A}^{-\top}$
Identidades para $\boldsymbol{\Sigma}$ (sugeridas en ej. 20):
Si $\Lambda=\boldsymbol{\Sigma}^{-1}$: $\nabla_\Lambda\ln|\Lambda|=\Lambda^{-\top}=\boldsymbol{\Sigma}^\top$
$\nabla_\Lambda\mathrm{tr}(\Lambda\mathbf{S})=\mathbf{S}^\top$
Proyección ortogonal: $\mathbf{P}=\hat{\mathbf{v}}\hat{\mathbf{v}}^\top$ (rango 1, idempotente)
$\|\mathbf{x}-\mathbf{P}\mathbf{x}\|^2=\|\mathbf{x}\|^2-(\mathbf{x}^\top\hat{\mathbf{v}})^2$
Matriz ortogonal R: $\mathbf{R}^\top\mathbf{R}=\mathbf{I}$, $|\det\mathbf{R}|=1$
$\|\mathbf{R}\mathbf{x}\|^2=\mathbf{x}^\top\mathbf{R}^\top\mathbf{R}\mathbf{x}=\|\mathbf{x}\|^2$ (isometría)
Autovalores y traza/det:
$\mathrm{tr}(\mathbf{A})=\sum_i\lambda_i$; $|\mathbf{A}|=\prod_i\lambda_i$
$\mathrm{tr}(\mathbf{AB})=\mathrm{tr}(\mathbf{BA})$ incluso si $\mathbf{A},\mathbf{B}$ no cuadradas
Restricción con Lagrange:
$\max_{\hat{\mathbf{v}}}\hat{\mathbf{v}}^\top\hat{\boldsymbol{\Sigma}}\hat{\mathbf{v}}$ s.t. $\|\hat{\mathbf{v}}\|=1$
$\Rightarrow\mathcal{L}=\hat{\mathbf{v}}^\top\hat{\boldsymbol{\Sigma}}\hat{\mathbf{v}}-\lambda(\hat{\mathbf{v}}^\top\hat{\mathbf{v}}-1)$
$\Rightarrow\hat{\boldsymbol{\Sigma}}\hat{\mathbf{v}}=\lambda\hat{\mathbf{v}}$ (autoproblema)
$\Rightarrow$ máx. varianza $=\lambda_1$ con $\hat{\mathbf{v}}=\mathbf{u}_1$
Gaussiana multivariada $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu},\boldsymbol{\Sigma})$:
$$\frac{1}{(2\pi)^{D/2}|\boldsymbol{\Sigma}|^{1/2}}\exp\Big(-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\Big)$$

Demo: EM-GMM - $\mathcal{Q}$ y concavidad

E-step — $\mathcal{Q}(\mathbf{w},\mathbf{w}^{\mathrm{old}})$:
$p(\mathbf{X},\mathbf{Z}|\mathbf{w})=\prod_n\prod_k[\pi_k\mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k)]^{z_{nk}}$
$\ln p(\mathbf{X},\mathbf{Z}|\mathbf{w})=$
$\sum_n\sum_k z_{nk}[\ln\pi_k+\ln\mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k)]$
$\mathcal{Q}=\sum_n\sum_k r_{nk}[\ln\pi_k-\frac{1}{2}\ln|\boldsymbol{\Sigma}_k|-$
$\frac{1}{2}(\mathbf{x}_n-\boldsymbol{\mu}_k)^\top\boldsymbol{\Sigma}_k^{-1}(\mathbf{x}_n-\boldsymbol{\mu}_k)]$
M-step para $\boldsymbol{\mu}_k$: $\nabla_{\boldsymbol{\mu}_k}\mathcal{Q}=0$
$\sum_n r_{nk}\boldsymbol{\Sigma}_k^{-1}(\mathbf{x}_n-\boldsymbol{\mu}_k)=0\Rightarrow\hat{\boldsymbol{\mu}}_k=\frac{\sum_n r_{nk}\mathbf{x}_n}{N_k}$
M-step para $\boldsymbol{\Sigma}_k$: $\nabla_{\Lambda_k}\mathcal{Q}=0$ con $\Lambda_k=\boldsymbol{\Sigma}_k^{-1}$:
$\frac{N_k}{2}\boldsymbol{\Sigma}_k-\frac{1}{2}\sum_n r_{nk}(\mathbf{x}_n-\hat{\boldsymbol{\mu}}_k)(\mathbf{x}_n-\hat{\boldsymbol{\mu}}_k)^\top=0$
$\Rightarrow\hat{\boldsymbol{\Sigma}}_k=\frac{1}{N_k}\sum_n r_{nk}(\mathbf{x}_n-\hat{\boldsymbol{\mu}}_k)(\mathbf{x}_n-\hat{\boldsymbol{\mu}}_k)^\top$
M-step para π_k (Lagrange $\sum\pi_k=1$):
$\nabla_{\pi_k}[\mathcal{Q}+\nu(\sum\pi_k-1)]=0\Rightarrow\hat{\pi}_k=\frac{N_k}{N}$
Concavidad de $\mathcal{Q}$: es una suma de $\ln\mathcal{N}$ ponderados; $\ln\mathcal{N}$ es cóncavo en $(\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k)\Rightarrow\mathcal{Q}$ cóncava $\Rightarrow$ M-step es único. $\square$

Demo: PCA minimiza error de recon.

Datos centrados ($\hat{\boldsymbol{\mu}}=0$). $\tilde{x}_i=z_i\hat{\mathbf{v}}$, $z_i=\mathbf{x}_i^\top\hat{\mathbf{v}}$.
$\mathrm{MSE}=\frac{1}{N}\sum_i\|\mathbf{x}_i-z_i\hat{\mathbf{v}}\|^2$
$=\frac{1}{N}\sum_i\|\mathbf{x}_i\|^2-\frac{1}{N}\sum_i(\mathbf{x}_i^\top\hat{\mathbf{v}})^2$
$=\mathrm{const}-\hat{\mathbf{v}}^\top\left(\frac{1}{N}\mathbf{X}^\top\mathbf{X}\right)\hat{\mathbf{v}}$
$\Rightarrow$ minimizar MSE $\equiv$ maximizar $\hat{\mathbf{v}}^\top\hat{\boldsymbol{\Sigma}}\hat{\mathbf{v}}$.
Con restricción $\|\hat{\mathbf{v}}\|=1$ (Lagrange):
$\hat{\boldsymbol{\Sigma}}\hat{\mathbf{v}}=\lambda\hat{\mathbf{v}}$; la varianza $=\lambda$.
$\Rightarrow\hat{\mathbf{v}}^*=\mathbf{u}_1$ (autovector de $\lambda_{\max}$).

Demo: GMM: $+K\implies$ log-lik no decrece

Sea $\mathbf{w}_K^*$ óptimo para K clusters. Construir modelo con $K+1$ clusters tomando $\mathbf{w}_{K+1}$ con $\pi_{K+1}=0$.
$\Rightarrow p_{K+1}(\mathbf{X}|\mathbf{w}_{K+1})=p_K(\mathbf{X}|\mathbf{w}_K^*)$
$\Rightarrow\ln p(\mathbf{X}|\mathbf{w}_{K+1}^*)\geq\ln p(\mathbf{X}|\mathbf{w}_{K+1})=\ln p(\mathbf{X}|\mathbf{w}_K^*)$